



Cohort canonical evaluation — 30 agents

idea-01-lateral-attribution · 534 held-out segments · V0
yaw 0.01627 rad/s, CTE 254.26 m · generated
2026-06-04 12:39

- **Executive summary**

AGENTS (OK / TOTAL)

30 / 30

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

14.26s

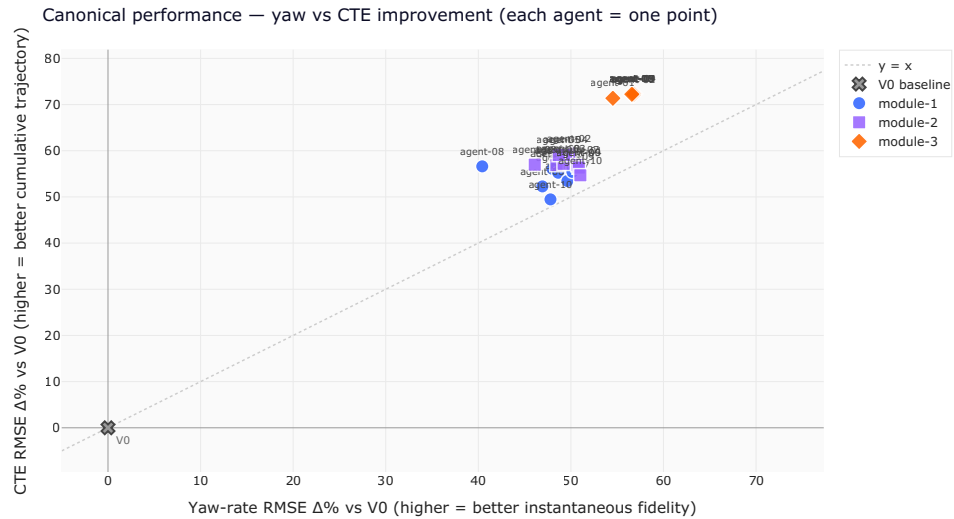
Best yaw: m3-agent-04 **+56.8%** · Best CTE: m3-agent-04

+72.3%

Winning both KPIs $\geq +30\%$ (30 agents): m3-agent-04, m3-agent-09, m3-agent-02, m3-agent-03, m3-agent-05, m3-agent-06, m3-agent-07, m3-agent-08, m3-agent-10, m3-agent-01, m2-agent-10, m1-agent-01, m2-agent-09, m2-agent-06, m2-agent-07, m1-agent-09, m2-agent-02, m1-agent-07, m2-agent-04, m2-agent-03, m2-agent-05, m1-agent-02, m1-agent-05, m1-agent-03, m2-agent-01, m1-agent-04, m1-agent-10, m1-agent-06, m2-agent-08, m1-agent-08

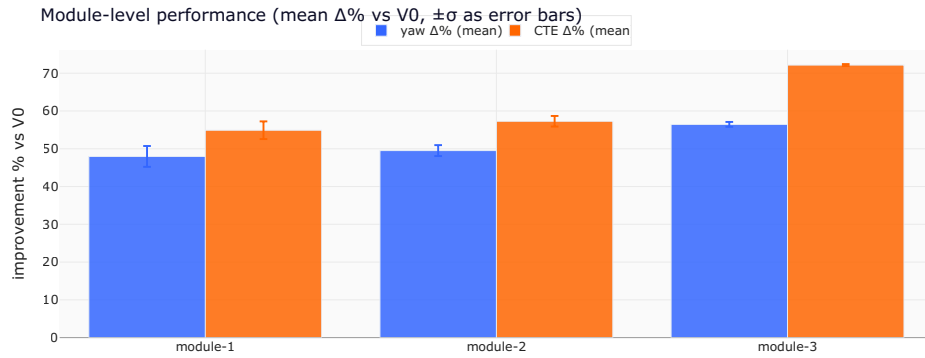
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$ (mean $\pm \sigma$)
module-1	10/10	+48.0% $\pm$ 2.8%	+54.9% $\pm$ 2.8%
module-2	10/10	+49.5% $\pm$ 1.4%	+57.3% $\pm$ 1.4%
module-3	10/10	+56.5% $\pm$ 0.6%	+72.2% $\pm$ 0.6%

• 2b. Token expenditure per agent

Sourced from each agent's Claude Code subagent transcript
 (~/.claude/projects/<proj>/*/subagents/agent-*.jsonl).
 Tokens summed across every assistant turn of the latest-
 mtime run per agent.

COHORT TOTAL

90.46M

MEDIAN / AGENT

2.67M

MEDIAN TURNS

63

AGENTS W/ TRANSCRIPT

30

By family

family	n	total tokens	median / agent	median turns
module-1	10	23.38M	2.38M	62
module-2	10	25.69M	2.24M	54
module-3	10	41.38M	3.50M	67

Per agent (sorted by total)

agent	family	turns	total	input	output
m3-agent-06	module-3	107	8.18M	132	23.6k
m3-agent-01	module-3	94	6.78M	109	28.6k
	module-3	106	6.03M	146	14.9k

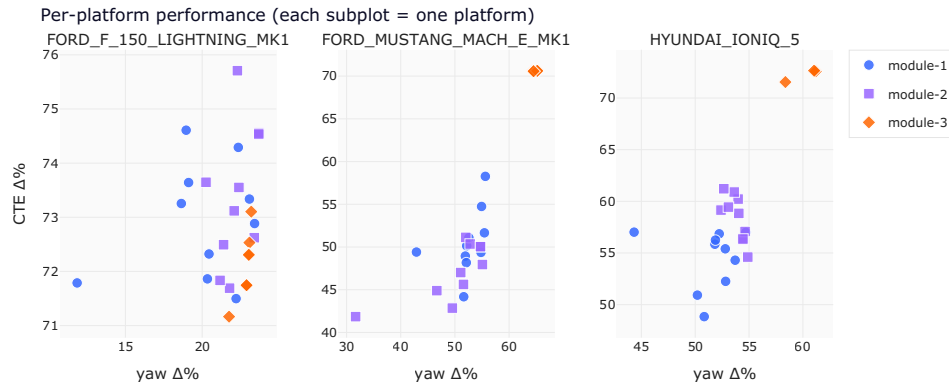
agent	family	turns	total	input	output
m3-agent-05					
m2-agent-05	module-2	83	4.45M	93	18.2k
m3-agent-03	module-3	72	3.96M	82	16.2k
m3-agent-08	module-3	68	3.84M	83	13.8k
m2-agent-08	module-2	76	3.73M	96	13.2k
m2-agent-07	module-2	69	3.61M	79	14.2k
m1-agent-09	module-1	79	3.54M	106	19.8k
m3-agent-07	module-3	66	3.16M	76	12.7k
m3-agent-09	module-3	61	3.02M	71	7.6k
m1-agent-08	module-1	74	2.96M	82	18.4k
m3-agent-10	module-3	65	2.86M	75	12.9k
m3-agent-02	module-3	61	2.83M	71	13.1k
	module-2	68	2.68M	78	16.0k

agent	family	turns	total	input	output
m2-agent-02					
m1-agent-02	module-1	66	2.66M	78	21.0k
m1-agent-10	module-1	64	2.45M	76	18.6k
m1-agent-07	module-1	62	2.41M	70	22.7k
m2-agent-03	module-2	52	2.35M	67	10.3k
m1-agent-04	module-1	63	2.34M	71	17.6k
m1-agent-05	module-1	62	2.25M	74	16.7k
m2-agent-09	module-2	55	2.14M	65	14.3k
m2-agent-10	module-2	53	2.11M	68	11.5k
m1-agent-01	module-1	53	1.88M	65	17.3k
m2-agent-04	module-2	45	1.83M	55	9.7k
m1-agent-06	module-1	54	1.73M	66	15.2k
	module-2	43	1.66M	53	13.7k

agent	family	turns	total	input	output
m2-agent-06					
m1-agent-03	module-1	38	1.17M	46	9.2k
m2-agent-01	module-2	41	1.13M	51	8.9k
m3-agent-04	module-3	26	740.2k	41	2.9k

- ### 3. Per-platform breakdown

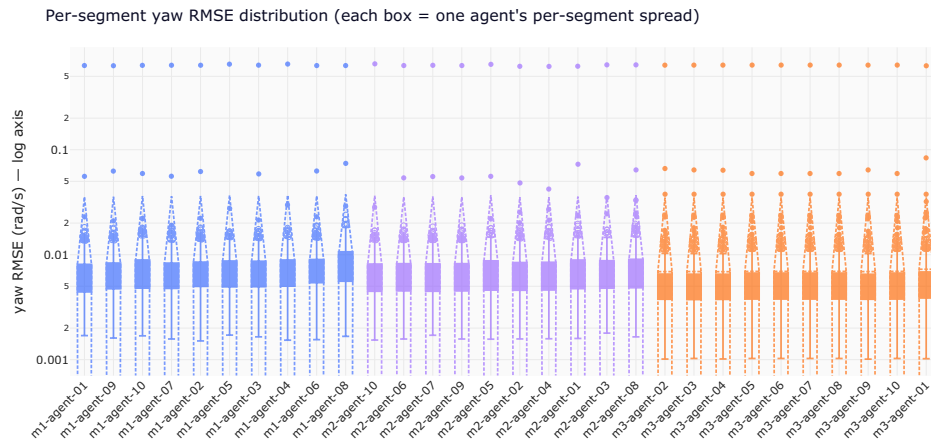
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw $\Delta\%$ (mean $\pm \sigma$)
FORD_F_150_LIGHTNING_MK1	30	+21.7% $\pm$ 2.3%
FORD_MUSTANG_MACH_E_MK1	30	+55.7% $\pm$ 7.8%
HYUNDAI_IONIQ_5	30	+55.4% $\pm$ 4.3%

- 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status	yaw V0	yaw final
m1-agent-01	module-1	ok	0.016274	0.007971
m1-agent-02	module-1	ok	0.016274	0.008358
m1-agent-03	module-1	ok	0.016274	0.008381
	module-1	ok	0.016274	0.008461

agent	family	status	yaw V0	yaw final	
m1-agent-04					
m1-agent-05	module-1	ok	0.016274	0.008360	
m1-agent-06	module-1	ok	0.016274	0.008638	
m1-agent-07	module-1	ok	0.016274	0.008199	
m1-agent-08	module-1	ok	0.016274	0.009695	
m1-agent-09	module-1	ok	0.016274	0.008111	
m1-agent-10	module-1	ok	0.016274	0.008496	
m2-agent-01	module-2	ok	0.016274	0.008391	
m2-agent-02	module-2	ok	0.016274	0.008167	
m2-agent-03	module-2	ok	0.016274	0.008264	
m2-agent-04	module-2	ok	0.016274	0.008224	
m2-agent-05	module-2	ok	0.016274	0.008350	
	module-2	ok	0.016274	0.007997	

agent	family	status	yaw V0	yaw final
m2-agent-06				
m2-agent-07	module-2	ok	0.016274	0.008023
m2-agent-08	module-2	ok	0.016274	0.008772
m2-agent-09	module-2	ok	0.016274	0.007997
m2-agent-10	module-2	ok	0.016274	0.007970
m3-agent-01	module-3	ok	0.016274	0.007399
m3-agent-02	module-3	ok	0.016274	0.007038
m3-agent-03	module-3	ok	0.016274	0.007045
m3-agent-04	module-3	ok	0.016274	0.007026
m3-agent-05	module-3	ok	0.016274	0.007065
m3-agent-06	module-3	ok	0.016274	0.007065
m3-agent-07	module-3	ok	0.016274	0.007065
	module-3	ok	0.016274	0.007065

agent	family	status	yaw V0	yaw final
m3-agent-08				
m3-agent-09	module-3	ok	0.016274	0.007032
m3-agent-10	module-3	ok	0.016274	0.007065

• 6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
```

```
    "L": 3.096229822470566,  
    "K_us": 0.003695564361835472,  
    "bias_rad": -0.0002960976826300522  
  },  
  "TESLA_MODEL_3": {  
    "L": 2.875,  
    "K_us": 0.0,  
    "bias_rad": 0.0  
  }  
}
```

m1-agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "a": -0.004329037631466362,  
    "b": 0.9177897626357783,  
    "c": -0.00044178131403551616  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "a": 0.0002206757945255914,  
    "b": 1.1513559927303039,  
    "c": -0.0005583363495612761  
  },  
  "HYUNDAI_IONIQ_5": {  
    "a": 0.0020222989265629857,  
    "b": 0.9120045284217976,  
    "c": -0.00048449411444108955  
  },  
  "TESLA_MODEL_3": {  
    "a": 0.0002206757945255914,  
    "b": 1.1513559927303039,  
    "c": -0.0005583363495612761  
  },  
}
```

```
"_default": {  
  "a": 0.0,  
  "b": 1.0,  
  "c": 0.0  
}  
}
```

m1-agent-06

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "best": "M2_understeer",  
    "m1": {  
      "a": 1.0870612234849137  
    },  
    "m2": {  
      "L_eff": 2.4763030468069056,  
      "K_us": 0.0024395167795073017  
    },  
    "m3": {  
      "a": 0.9751496691433119,  
      "b": 0.23839497273290036,  
      "c": 0.0004082870644824788  
    },  
    "val_rmse": {
```

```
    "V0": 0.0267999381028469,  
    "M1": 0.027439581036525294,  
    "M2": 0.026420879339395156,  
    "M3": 0.031903168689021696  
  },  
  "train_rmse": {  
    "V0": 0.012685628695945105,  
    "M1": 0.011389013310887795,  
    "M2": 0.008745921115146504,  
    "M3": 0.010386863485176999  
  }  
},  
"FORD_F_150_LIGHTNING_MK1": {  
  "L": 3.7,  
  "best": "M2_understeer",  
  "m1": {  
    "a": 0.8569324797501493  
  },  
  "m2": {  
    "L_eff": 3.882198880475702,  
    "K_us": 0.003439279945738158  
  },  
  "m3": {  
    "a": 0.8772240685582882,  
    "b": -0.031353830652592465,  
    "c": -0.0036354381262862774  
  },  
  "val_rmse": {  
    "V0": 0.01591104278075153,  
    "M1": 0.011141896721881754,  
    "M2": 0.0073696198273352906,  
    "M3": 0.0104650790814036  
  },  
  "train_rmse": {  
    "V0": 0.02018452928972004,
```


m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```

m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.",
    "passthrough": true
  }
}
```

m2-agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.941941906538264,
    "K_us": 0.0037788233688761494,
    "b": -0.005450661568006636,
    "tau": -0.08207658150714123
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5622977171066155,
    "K_us": 0.002745633390344498,
    "b": 0.00039676132180812505,
    "tau": -0.02283428861943423
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.0387877407606987,
    "K_us": 0.004732031059157526,
    "b": 0.0019402727789656954,
    "tau": -0.038122141286844304
  },
  "TESLA_MODEL_3": null
}
```

m2-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.96394551474604,
    "Kus": 0.003414866921617812,
    "bias": -0.005292773687032753,
    "tau": -0.06157679065853846
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.569483805164948,
    "Kus": 0.0025788290081397765,
    "bias": 0.0011230859100703557,
    "tau": -0.04709908110775255
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.008377713685593,
    "Kus": 0.005133658670161866,
    "bias": 0.0019487320168033527,
    "tau": -0.04867295794401198
  },
  "TESLA_MODEL_3": {
    "dummy": 0.0
  }
}
```

m2-agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.818759956786134,
    "Kus": 0.0039432563422251285,
    "tau": -0.05861405092290705,
    "bias": -0.004437268567086911,

```

```
    "rmse_fit": 0.0054979371675811595,  
    "n": 430990,  
    "converged": true  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "platform": "FORD_MUSTANG_MACH_E_MK1",  
    "L": 2.4997194805464074,  
    "Kus": 0.0024024802542751437,  
    "tau": -0.06630813663570458,  
    "bias": 0.00022227158439854945,  
    "rmse_fit": 0.00894166094428278,  
    "n": 615175,  
    "converged": true  
  },  
  "HYUNDAI_IONIQ_5": {  
    "platform": "HYUNDAI_IONIQ_5",  
    "L": 3.100074393615448,  
    "Kus": 0.0035463766130646316,  
    "tau": -0.05358525493540066,  
    "bias": 0.001966636983251082,  
    "rmse_fit": 0.008266374104865987,  
    "n": 2042270,  
    "converged": true  
  }  
}
```

m2-agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {
```

```
"L": 3.7,
"K_us": 0.004622784347956018,
"tau": -0.05460138412592134,
"bias": -0.004482561615678892,
"train_mse": 3.3520085701320475e-05,
"train_rmse": 0.005789653331704799,
"n_samples": 430990,
"n_segments": 164,
"converged": true,
"message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
},
"FORD_MUSTANG_MACH_E_MK1": {
  "L": 2.984,
  "K_us": 0.0008351228965168481,
  "tau": -0.09459794094766563,
  "bias": 0.00054055263346224,
  "train_mse": 0.00017605613840148047,
  "train_rmse": 0.01326861478834473,
  "n_samples": 615175,
  "n_segments": 232,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
},
"HYUNDAI_IONIQ_5": {
  "L": 2.95,
  "K_us": 0.004646731366208564,
  "tau": -0.04946605447751251,
  "bias": 0.0018472769568295111,
  "train_mse": 7.578893405982132e-05,
  "train_rmse": 0.008705684008727937,
  "n_samples": 2042270,
  "n_segments": 785,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
}
}
```

m2-agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "s_d": 0.9300245172021879,
    "c_d": 0.347294150023435,
    "tau_d": -0.05773764830098028,
    "K_us": 0.003240471803333766,
    "b": -0.004299834214218103,
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "s_d": 1.1508056993666387,
    "c_d": 0.7498585082419895,
    "tau_d": -0.07607250144234215,
    "K_us": 0.0022785419555576775,
    "b": 0.0003167396785048121,
    "L": 2.984
  },
  "HYUNDAI_IONIQ_5": {
    "s_d": 0.9422644146648296,
    "c_d": 0.3465308670029774,
    "tau_d": -0.050503079814885346,
    "K_us": 0.002862544790766583,
    "b": 0.0022556662651586025,
    "L": 3.0
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```

m3-agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8599716540800916,
```

```

    "L_eff": 3.260802128118046,
    "K_us": 0.0033605485538025595,
    "tau": 0.058798336746219135,
    "delta0": 0.0012445223286377857,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8948751688442338,
    "L_eff": 2.218474888909999,
    "K_us": 0.002002066992546276,
    "tau": 0.06321825538415474,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.928157495232038,
    "L_eff": 2.890194195032293,
    "K_us": 0.002716802199963435,
    "tau": 0.051288867659338114,
    "delta0": 0.0005262026839657872,
    "use_per_segment_delta0": true
  }
}

```

m3-agent-04

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8627559173927216,
    "L_eff": 3.2661642837279934,
    "K_us": 0.003399629365556295,
    "tau": 0.05764217451751205,
    "delta0": 0.0011797692301086372,
    "delta0_fallback": 0.0011797692301086372,
    "use_per_segment_delta0": false,
    "train_rmse": 0.005285188367801501,
    "dev_rmse": 0.006580546885659461,
    "bias_spread_std": null
  }
}

```



```
},
"FORD_MUSTANG_MACH_E_MK1": {
  "g": 1.2845766366025777,
  "L_eff": 3.185078856835661,
  "K_us": 0.002783912493950955,
  "tau": 0.06245560987504152,
  "delta0": -0.0019790637549735985,
  "delta0_fallback": -0.0019790637549735985,
  "use_per_segment_delta0": true,
  "train_rmse": 0.008444302053948288,
  "dev_rmse": 0.008286227301504138,
  "bias_spread_std": null
},
"HYUNDAI_IONIQ_5": {
  "g": 0.9454031576502736,
  "L_eff": 2.9345636591691626,
  "K_us": 0.0028198034203513055,
  "tau": 0.0507163687871907,
  "delta0": 0.0004041780050478772,
  "delta0_fallback": 0.0004041780050478772,
  "use_per_segment_delta0": true,
  "train_rmse": 0.007930751971921808,
  "dev_rmse": 0.00660747263955304,
  "bias_spread_std": null
}
}
```

m3-agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
}
```

```
"FORD_MUSTANG_MACH_E_MK1": {
  "use_per_segment_delta0": true,
  "delta0_fallback": -0.0001,
  "g": 0.891,
  "L_eff": 2.22,
  "K_us": 0.0015,
  "tau": 0.069
},
"HYUNDAI_IONIQ_5": {
  "use_per_segment_delta0": true,
  "delta0_fallback": 0.0,
  "g": 0.938,
  "L_eff": 2.887,
  "K_us": 0.00289,
  "tau": 0.062
}
}
```

m3-agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
```

```
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```

m3-agent-10

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "use_per_segment_delta0": false,  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```

```
"L_eff": 2.887,  
"K_us": 0.00289,  
"tau": 0.062  
}  
}
```

● 6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

Clean cohort: no agent referenced truth-derived column names inside their `predict()` body.

- ## 7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	30	0
has_manifest_json	30	0
manifest_parsable	30	0
manifest_declares_predict_callable	30	0
manifest_declares_platform_support	30	0
has_predict_py	30	0
has_coeffs_json	29	1
has_report	16	14

- ## 8. Worst-of-cohort

Lowest yaw $\Delta\%$

- m1-agent-08
(+40.4%)
- m2-agent-08
(+46.1%)
- m1-agent-06
(+46.9%)

Lowest CTE $\Delta\%$

- m1-agent-10
(+49.5%)
- m1-agent-06
(+52.3%)
- m1-agent-07
(+53.4%)